

Program 7

Step 1: Start Minikube

```
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs$ minikube start --driver=docker
🐳 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...

🚨 Docker is nearly out of disk space, which may cause deployments to fail! (87% of capacity)
. You can pass '--force' to skip this check.
💡 Suggestion:

Try one or more of the following to free up space on the device:

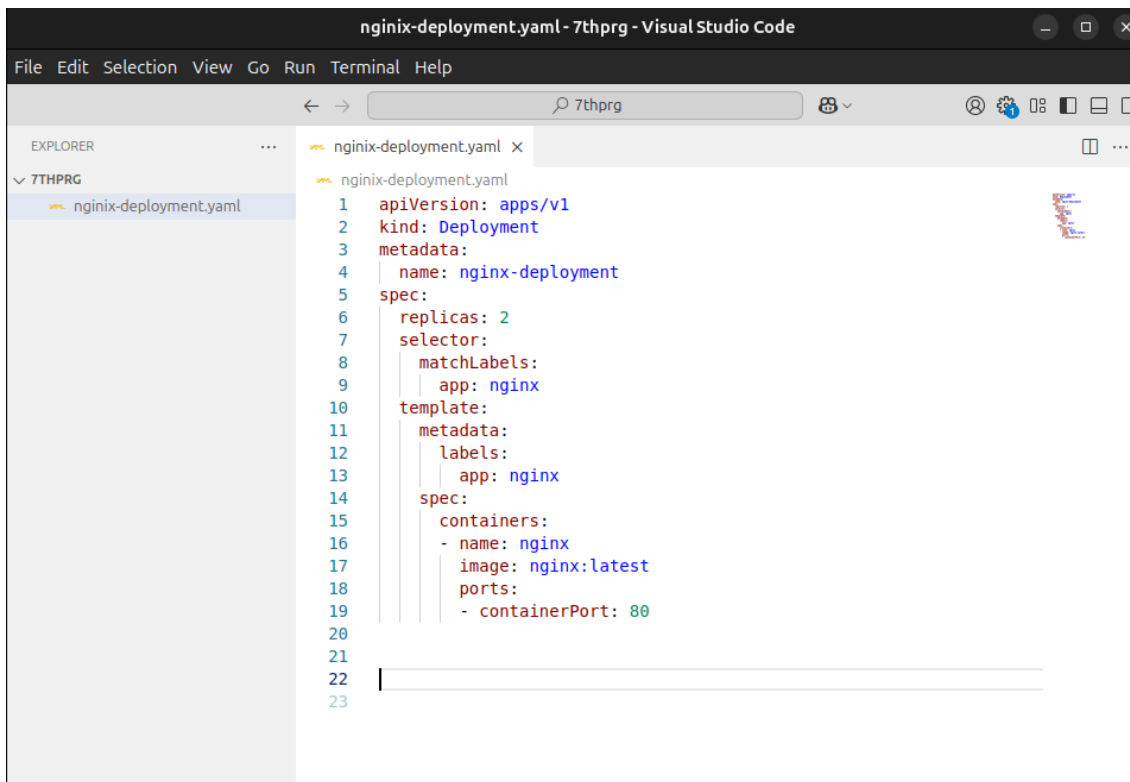
1. Run "docker system prune" to remove unused Docker data (optionally with "-a")
2. Increase the storage allocated to Docker for Desktop by clicking on:
   Docker icon > Preferences > Resources > Disk Image Size
3. Run "minikube ssh -- docker system prune" if using the Docker container runtime
```

Step 2: Check Minikube Status

Step 3: Verify Running Services

```
1rv24mc077_prateek_bhandari@prateek: ~/dockerprgs/7thprg
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs$ mkdir 7thprg
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs$ cd 7thprg/
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/7thprg$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/7thprg$ touch nginx-deployment.yaml
```

Step 5: Create Deployment YAML

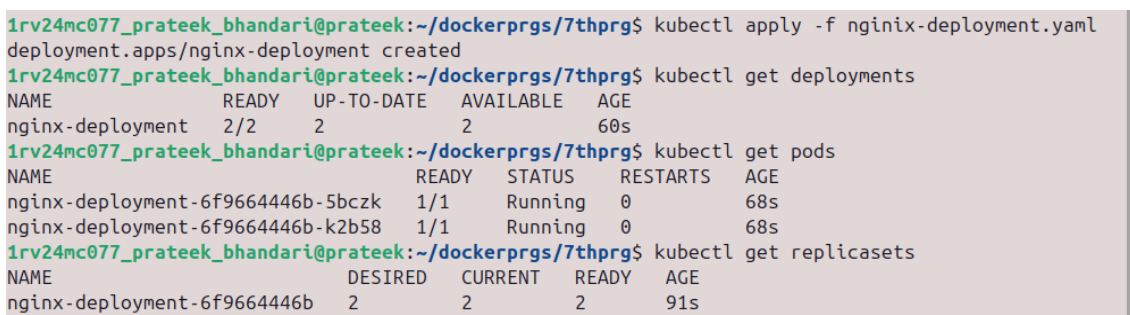


The screenshot shows the Visual Studio Code editor with a file named `nginx-deployment.yaml` open. The Explorer sidebar on the left shows the file structure with a folder named `7THPRG` containing the `nginx-deployment.yaml` file. The main editor area displays the following YAML content:

```
1 apiVersion: apps/v1
2 kind: Deployment
3 metadata:
4   name: nginx-deployment
5 spec:
6   replicas: 2
7   selector:
8     matchLabels:
9       app: nginx
10  template:
11    metadata:
12      labels:
13        app: nginx
14    spec:
15      containers:
16        - name: nginx
17          image: nginx:latest
18          ports:
19            - containerPort: 80
```

Step 6: Apply Deployment

Step 7: Check Status



The screenshot shows a terminal window with the following commands and output:

```
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/7thprg$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/7thprg$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    2/2     2            2           60s
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/7thprg$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-5bczk   1/1     Running   0           68s
nginx-deployment-6f9664446b-k2b58   1/1     Running   0           68s
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/7thprg$ kubectl get replicaset
NAME                                DESIRED   CURRENT   READY   AGE
nginx-deployment-6f9664446b         2         2         2       91s
```

Step 8: Expose Deployment as a Service

Step 9: Check Service

Step 10: Access the Web Application

```
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/7thprg$ kubectl expose deployment nginx-deployment \
--type=NodePort \
--port=80
service/nginx-deployment exposed
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/7thprg$ kubectl get services
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
kubernetes           ClusterIP   10.96.0.1     <none>       443/TCP          3m18s
nginx-deployment     NodePort    10.99.81.214  <none>       80:31261/TCP     45s
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/7thprg$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:31261

 Opening service default/nginx-deployment in default browser...

```
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/7thprg$ Opening in existing browser session.
```

Step 11: Delete All Pods

Step 12: Delete All Services

Step 13: Delete All Deployments

```
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/7thprg$ kubectl delete pods --all
pod "nginx-deployment-6f9664446b-5bczk" deleted from default namespace
pod "nginx-deployment-6f9664446b-k2b58" deleted from default namespace
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/7thprg$ kubectl delete svc --all
service "kubernetes" deleted from default namespace
service "nginx-deployment" deleted from default namespace
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/7thprg$ kubectl delete deployments --all
deployment.apps "nginx-deployment" deleted from default namespace
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/7thprg$
```